Phase 0: Data Extraction Pipeline

Vancomycin Pharmacokinetic Cohort from MIMIC-IV

Data source: MIMIC-IV Clinical Database (v3.1)

Objective: Build a clean, analysis-ready dataset for vancomycin pharmacokinetic modelling by extracting and integrating drug concentrations, dose administrations, laboratory values, and clinical covariates from the raw electronic health record.

Contents

1 Dataset and Objective **2**
1.1 MIMIC-IV . . . 2
1.2 Clinical Objective . . . 2

2 Database Structure and Identifiers **2**

3 Item Identifier Resolution **3**

4 Vancomycin Concentration Extraction **4**
4.1 Source Decision . . . 4
4.2 Extraction Strategy . . . 4
4.3 Results . . . 4

5 Vancomycin Dose Extraction **5**
5.1 Source Decision . . . 5
5.2 Extraction Strategy . . . 5
5.3 Results . . . 6

6 Laboratory Value Extraction **6**
6.1 Cohort Definition . . . 6
6.2 Event Type Mapping . . . 6
6.3 Results . . . 7

7 Clinical Covariates **7**
7.1 Demographics . . . 7
7.2 Ventilation Flags . . . 8
7.3 Dialysis Flags . . . 8
7.4 Vasopressor Flag . . . 8
7.5 Final Covariates Table . . . 9

8 Final Cohort and Output Files **9**
8.1 Cohort Construction . . . 9
8.2 Output Files . . . 9
8.3 Unified Event Schema . . . 9

1 Dataset and Objective

1.1 MIMIC-IV

MIMIC-IV (Medical Information Mart for Intensive Care, version 3.1) is a large, de-identified electronic health record database containing data from hospital admissions of 364 627 unique patients admitted to the Beth Israel Deaconess Medical Center. The database is organised into two main modules:

- **Hospital module** : covers the full hospitalisation from admission to discharge, including laboratory results and prescriptions.
- **ICU module** : covers intensive care unit stays specifically, including charted vitals, infusion pump records, and procedure events.

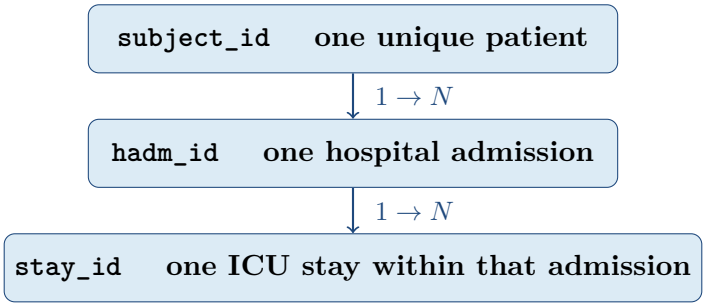
1.2 Clinical Objective

The goal of this project is to build a vancomycin dosing recommendation system given a patient's dosing history, blood concentration measurements, and clinical covariates (age, sex, weight, height...).

Phase 0 is entirely concerned with data extraction: producing four clean, analysis-ready files that subsequent modelling phases will consume.

2 Database Structure and Identifiers

MIMIC-IV organises records across three nested levels, each representing a genuinely different clinical event with its own timestamps and clinical state:



A single patient (`subject_id`) can have multiple hospital admissions (`hadm_id`) over their lifetime, and within one admission may have multiple ICU stays (`stay_id`). All timing computations use `stay_id` as the anchor, with elapsed time measured in hours from ICU admission (`intime`).

MIMIC does not store drug names or measurement names as strings in the large event tables, instead it stores them as integer codes called `itemid`. Two dictionaries translate these codes into human-readable labels:

- `d_items` — for ICU chart and pump events
- `d_labitems` — for hospital laboratory events

3 Item Identifier Resolution

Before any data extraction, each clinical variable of interest was resolved to its numeric `itemid`(s) by querying the dictionaries. All resolved identifiers are declared as named constants at the top of the extraction pipeline and referenced by name throughout. Table 1 summarises the complete mapping.

Table 1: Resolved `itemid` constants used in the extraction pipeline.

Constant	Item IDs	Dictionary	Scope
`VANCO_DOSE_ITEMIDS`	225798	`d_items`	ICU pump (inputc
`VANCO_LAB_ITEMID`	51009	`d_labitems`	Lab level (labeven
`VANCO_LEVEL_ITEMIDS`	227453, 227454, 227455	`d_items`	Bedside level (cha
`WEIGHT_ITEMIDS`	226512	`d_items`	Admission weight
`HEIGHT_ITEMIDS`	226730	`d_items`	Height (cm)
`CREATININE_ITEMIDS`	50912, 52546	`d_labitems`	Blood creatinine
`BUN_ITEMIDS`	51006, 52647	`d_labitems`	Blood urea nitrog
`ALBUMIN_ITEMIDS`	50862, 52022, 53085, 53138	`d_labitems`	Blood albumin
`ALT_ITEMIDS`	50861	`d_labitems`	Alanine aminotrar
`AST_ITEMIDS`	50878	`d_labitems`	Aspartate aminotr
`VASOPRESSOR_ITEMIDS`	221906, 221289, 221662, 222315, 221749	`d_items`	Vasopressors
`MECH_VENT_ITEMIDS`	225792, 225794	`d_items`	Mechanical ventila
`DIALYSIS_ITEMIDS`	225441, 225802, 225803, 225809, 225805, 225955	`d_items`	All dialysis modal

Several variables had multiple valid item IDs — for example, creatinine appears under both `50912` and `52546` in `d_labitems`. Only blood-source entries were retained in all cases; urine, CSF, ascites, and other fluid measurements were explicitly excluded.

For the vancomycin level items in `chartevents`, three timing variants exist, with the following record counts:

Item ID	Label	Records
227455	Trough (drawn immediately before next dose)	20 092
227454	Random (drawn at an arbitrary time)	20 020
227453	Peak (drawn shortly after infusion ends)	18

Peak monitoring (227453) is nearly absent (18 records). A fourth family of "Z-prefixed" items (225697, 226064, 226065) was queried and confirmed to contain **zero records** — these identifiers exist in `d_items` and are excluded entirely.

4 Vancomycin Concentration Extraction

4.1 Source Decision

Vancomycin blood concentrations exist in two separate tables in MIMIC-IV:

	labevents	**chartevents**
Origin	Central laboratory system	Nurse/bedside documentation
Nature	Ground-truth measurement	Human re-entry of the lab result
Risk	None	Rounding errors, transcription typos
Coverage	$N = 23\,952$ stays	Partial overlap

The fundamental problem with `chartevents` is that it is often a *copy*: when the laboratory reports a result (e.g. 14.3 mg/L), a nurse manually re-enters that value into the bedside chart, where it may be rounded (14.0 mg/L) or miskeyed and timestamped at a slightly different time. `labevents` item `51009` represents the original, unmodified laboratory measurement and is the authoritative source.

4.2 Extraction Strategy

`labevents` was used as the primary source. However, 75 ICU stays had vancomycin level records in `chartevents` but no corresponding entry in `labevents` — these were patients whose measurements were only documented at the bedside. To avoid losing these stays entirely, their `chartevents` records were extracted separately and appended to the main dataset.

The merge logic ensured no stay was counted twice: chartevents records were only included for stays with *no* matching `labevents` entry.

All concentration records were assigned `event_type` = 1 and elapsed time was computed as hours since ICU admission (`intime`).

4.3 Results

Table 2: Vancomycin concentration extraction results (`df_concentration`).

Source	**Unique stays**	**Total records**
`labevents` (primary)	23 952	83 474
`chartevents` patch (75 stays not in labevents)	75	201
Combined (`df_concentration`)	24 027	83 675

5 Vancomycin Dose Extraction

5.1 Source Decision

Vancomycin dose data exists in two complementary tables, each with distinct strengths and weaknesses:

	prescriptions	**inputevents**
What it records	Physician order	ICU pump log
Timing	Order activation time	Actual pump start time
Dose in mg	✓ 97% of rows	× 96.6% store "1 dose"
Precise timestamp	×	✓
Patient weight	×	✓
Coverage	All inpatients	ICU only

Neither table alone is sufficient: `prescriptions` provides the dose quantity in milligrams but uses order times rather than actual administration times; `inputevents` provides precise pump timestamps and patient weight but stores a unit count ("1 dose") rather than milligrams in 96.6% of rows.

5.2 Extraction Strategy

The two tables were joined using a time-window match: each pump event (`inputevents`) was matched to the active physician order (`prescriptions`) whose window covered the administration time (`admin_time` $\in$ [`order_start`, `order_stop`]). When multiple orders overlapped, the most recently activated order was selected via a `ROW_NUMBER` window function ordered by `order_start` descending.

Filters applied:

- `inputevents`: only `statusdescription` = `FinishedRunning` and `amount` > 0
- `prescriptions`: only `dose_unit_rx` = `mg` and `TRY_CAST(dose_val_rx AS DOUBLE)` $>$ 0

Of the 85 684 unique admissions with a vancomycin prescription, 28 657 (33.5%) also appeared in `inputevents`. The remaining 56 833 were non-ICU or ward patients who had no pump record in the ICU system.

Elapsed time was computed from ICU `intime` by joining through `icustays` on `stay_id`. All dose records were assigned `event_type` = 0.

5.3 Results

Table 3: Vancomycin dose extraction results (`df_dose`).

Metric	**Value**
Total dose events	121 672
Unique hospital admissions	27 180
Unique patients	23 454
Dose (mg) distribution	
Minimum	100 mg
5th percentile	750 mg
Median (50th percentile)	1000 mg
95th percentile	1500 mg
Maximum	2000 mg

6 Laboratory Value Extraction

6.1 Cohort Definition

Before extracting laboratory values, a cohort was defined as the intersection of stays present in all three of the preceding datasets: those with at least one vancomycin concentration record *and* at least one vancomycin dose record. Stays lacking either were excluded, as they cannot support pharmacokinetic modelling.

Table 4: Cohort construction by intersection.

Dataset	**Unique stays**
`df_concentration`	24 027
`df_dose`	28 912
Intersection (both concentration + dose)	18 993

Laboratory values were then extracted exclusively for stays in this cohort, and only for measurements falling within the ICU stay window (`charttime` $\geq$ `intime` and $\leq$ `outtime`).

6.2 Event Type Mapping

Each laboratory variable was assigned a numeric `event_type` code, consistent with the coding used in the concentration and dose tables:

Table 5: `event_type` encoding scheme across all extraction outputs.

event_type	Variable	Source item IDs
0	Vancomycin dose (mg)	225798
1	Vancomycin concentration (mg/L)	51009; 227453, 227454, 227455
2	Creatinine (mg/dL)	50912, 52546
3	BUN (mg/dL)	51006, 52647
4	Albumin (g/dL)	50862, 52022, 53085, 53138
5	ALT (U/L)	50861
6	AST (U/L)	50878

6.3 Results

Table 6: Laboratory value extraction results per variable (`df_labs`).

Type	Variable	Stays	Records	P5	Median	P95	Unit
2	Creatinine	18 699	248 383	0.4	1.2	4.7	mg/dL
3	BUN	18 697	248 014	8.0	28.0	93.0	mg/dL
4	Albumin	11 107	36 245	1.8	2.7	3.9	g/dL
5	ALT	13 812	75 319	9.0	43.0	974.0	U/L
6	AST	13 921	76 231	15.0	59.0	1227.0	U/L
Total		18 705	684 192				

7 Clinical Covariates

Static per-stay clinical characteristics were extracted from the 16,460-stay final cohort (see Section 8) and merged into a single wide table. Covariates are grouped into four categories.

7.1 Demographics

Age (in years at hospital admission) and sex were extracted from the `patients` table, joined to `icustays` on `subject_id`. ICU length of stay was derived as LOS_{ICU} = (`outtime` − `intime`)/3600 in hours.

Table 7: Continuous covariate summary (final cohort, $N = 16\,460$ stays).

Variable	N	Mean	SD	Min	Median	Max
Age (years)	16 460	62.4	15.9	18	64	91
Weight (kg)	14 814	83.9	27.5	1.0	80	710
Height (cm)†	10 109	—	—	—	170	—
ICU LOS (hours)	16 460	194	214	3.1	120	3832

7.2 Ventilation Flags

Mechanical ventilation status was extracted from `procedureevents` as two binary indicator columns.

Table 8: Ventilation and dialysis flag prevalence in the final cohort.

Category	Column	N positive	%
Ventilation	`has_invasive_vent`	10 101	61.4
	`has_non_invasive_vent`	1019	6.2
Dialysis	`has_hd`	1219	7.4
	`has_crrt`	1591	9.7
	`has_pd`	49	0.3
	`has_other_dialysis`	0	0.0
Vasopressor	`has_vasopressor`	6232	37.86

7.3 Dialysis Flags

Four dialysis modalities were extracted from `procedureevents`: haemodialysis (HD), continuous renal replacement therapy (CRRT), peritoneal dialysis (PD), and other (SCUF). Together, CRRT (1591 stays, 9.7%) and HD (1219 stays, 7.4%) account for the bulk of renal replacement therapy in the cohort. **Clinical significance.** The distinction between modalities matters for vancomycin dosing because they differ fundamentally in how much drug they remove: HD delivers high-efficiency clearance in short sessions, whereas CRRT provides slow, continuous removal over days. A patient on CRRT requires a different dosing strategy than one on HD — treating them as a single "on dialysis" flag would obscure this difference.

7.4 Vasopressor Flag

Vasopressor administration was extracted from `inputevents` using five item IDs: norepinephrine (221906), epinephrine (221289), dopamine (221662), vasopressin (222315), and phenylephrine (221749). The resulting binary column `has_vasopressor` indicates whether the patient received any vasopressor at any point during their ICU stay.

Clinical significance. Vasopressors are drugs administered when a patient's blood pressure drops dangerously — typically in septic shock or cardiovascular collapse. Their presence is a marker of critical illness severity. For vancomycin pharmacokinetics specifically, vasopressor use reduces renal perfusion, which slows drug clearance, making it an important covariate in any population PK model.

7.5 Final Covariates Table

The five sub-extractions (demographics, anthropometrics, ventilation, dialysis, vasopressor) were merged on `stay_id` using left joins anchored to the demographics base. Missing flag values for stays with no procedure record were filled with 0. The final `covariates` table has shape 16,460 × 16.

8 Final Cohort and Output Files

8.1 Cohort Construction

The final analysis cohort was defined as the intersection of stays present in all three of the primary extraction outputs: concentrations, doses, and laboratory values. Stays missing any of these three data streams cannot support pharmacokinetic parameter estimation and were excluded.

Table 9: Final cohort construction by three-way intersection.

Dataset	Unique stays
`df_concentration`	24 027
`df_dose`	28 912
`df_labs`	18 705
Final cohort (intersection of all three)	16 460

8.2 Output Files

Four Parquet files were saved as the output of Phase 0. All subsequent modelling phases consume these files directly; no further SQL queries against the raw MIMIC-IV tables are required.

Table 10: Phase 0 output files.

File	Rows	Columns	Size	Content
`covariates.parquet`	16 460	16	835 KB	One row per ICU stay
`df_concentration.parquet`	83 675	4	1.0 MB	Timestamped vanco levels
`df_dose.parquet`	121 672	6	1.9 MB	Timestamped dose events
`df_labs.parquet`	684 192	4	3.5 MB	Timestamped lab values

8.3 Unified Event Schema

The three time-series files (`df_concentration`, `df_dose`, `df_labs`) share a common four-column schema, enabling them to be stacked into a single longitudinal event stream per stay for modelling:

Column	Type	Description
`stay_id`	integer	ICU stay identifier
`time_hours`	float	Hours elapsed since ICU admission
`valuenum`	float	Observed value (mg, mg/L, mg/dL, etc.)
`event_type`	integer	Event category code (see Table 5)